Terpene diversity, biosynthetic evolution, and antiparasitic evidence across *Artemisia* species

Comprehensive review and comparative-genomics framework

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Abstract

The genus *Artemisia* contains a chemically diverse collection of volatile terpenes, nonvolatile terpenoids, and sesquiterpene lactones. This review develops a source-backed framework for comparing that chemistry across species while avoiding a common error: treating an essential-oil profile, a pathway gene, or an in-vitro parasite assay as a genus-wide pharmacological conclusion. Across 133 registered sources, we synthesize monoterpenes, sesquiterpenes, diterpenes, triterpenes, and sesquiterpene lactones; evaluate biosynthetic compartmentalization and gene-family evolution; and connect candidate pathway changes to chemotype and antiparasitic evidence. The best-resolved pathway is artemisinin biosynthesis in *A. annua*, but recent *A. argyi* results illustrate why gene presence/absence claims require assembly- and annotation-aware reconciliation. Malaria and artemisinin form the primary validated case study, while wormwood (*A. absinthium*) provides a separate, preparation-specific veterinary vermifuge/anthelmintic case. Essential-oil, extract, whole-leaf, and purified-compound assays provide phenotype anchors across protozoa, helminths, and vectors, yet constituent attribution remains unresolved in most mixture studies. We propose a reproducible, phylogeny-aware research program combining voucher-linked metabolomics, secretory-tissue transcriptomics, comparative genomics, enzyme assays, and fractionated parasite testing.

Keywords: *Artemisia*; terpenes; sesquiterpene lactones; artemisinin; chemotype; comparative genomics; essential oil; antiparasitic activity

Introduction

Artemisia is a large genus with substantial ecological, morphological, ploidy, and phytochemical diversity. Its volatile chemistry is concentrated especially in leaves and flowers, but reported profiles vary with genotype, chemotype, locality, tissue, ontogeny, harvest, drying, and extraction. Consequently, “the terpene profile of *Artemisia*” is not a single biological object; the appropriate unit of comparison is a documented specimen or accession with analytical context.

This review has two aims. First, it organizes the major terpene classes reported from *Artemisia* and summarizes the biosynthetic logic connecting precursor supply, synthase diversification, oxidation, lactonization, storage, and regulation. Second, it evaluates whether comparative evolutionary genomics can connect that diversity to antiparasitic phenotypes without overstating causality.

The evidence extraction unit is a specimen, accession, or explicitly bounded experimental preparation. Representative records are provided in `chemotype-table.csv`; they are not species means. The review follows the protocol in `review-protocol.md` and the source identifiers in `sources.json`.

What previous reviews establish-and what remains open

Yes, *Artemisia* terpene and terpenoid reviews have been published. The principal precedents include a genus-level essential-oil review, a 2012-2017 essential-oil update, a dedicated sesquiterpene-lactone analysis and methods review, a review of edible species and selected sesquiterpene lactones, and a 2024 genus-wide phytochemistry, pharmacology, and toxicology review (Abad et al. 2012; Pandey and Singh 2017; Ivanescu et al. 2015; Trendafilova et al. 2021; Hussain et al. 2024). These reviews establish the breadth of reported chemistry and biological claims, but they generally do not provide one specimen-level, class-balanced, phylogeny-aware evidence model connecting terpene chemistry, evolutionary genomics, tissue context, antiparasitic stage, and safety. The present package is designed as an auditable update and integration, not as a claim that prior reviews were absent or noncomprehensive.

Literature-search and evidence strategy

The literature review was organized around five linked evidence domains: (i) compound occurrence and analytical identity; (ii) enzyme, pathway, and compartment evidence; (iii) genotype, population, tissue, developmental, and environmental variation; (iv) comparative genomics and gene-family evolution; and (v) antiparasitic phenotype, target, and safety evidence. Searches combined *Artemisia* taxon terms with terpene-class, analytical, biosynthetic, genomic, and parasite terms. The reproducible search strings, dates, database results, de-duplication outputs, and title/abstract decisions are archived in `search-log.md`, `search-results.json`, and the screening CSV files. The working corpus contains both primary studies and prior reviews, but mechanistic and quantitative statements are preferentially anchored to primary studies.

Evidence was graded by what an experiment actually established. Isolation with NMR or a matched analytical standard supports compound identity in the tested material; GC-MS library matching supports a lower-confidence assignment unless retention-index and standard data are also available. Heterologous enzyme assays support catalytic capacity under the assay conditions, not native flux. Transcript abundance and co-expression prioritize candidates but do not prove product identity. Parasite growth inhibition supports a preparation-specific phenotype; causal target claims require biochemical, structural, genetic, or activity-based target-engagement evidence. This hierarchy allows broad coverage without converting association into mechanism.

The review is intentionally class-balanced. Essential-oil surveys dominate the searchable literature, but the synthesis separately tracks nonvolatile sesquiterpene lactones, diterpenes, triterpenes, and cuticular metabolites so that analytical convenience does not become a biological conclusion. Likewise, *A. annua* is treated as the best-resolved reference rather than a proxy for the genus. Evidence from *A. argyi*, *A. absinthium*, *A. herba-alba*, *A. afra*, *A. campestris*, and other species is retained with its own voucher, population, tissue, and preparation boundaries wherever those data are reported.

Chemical scope and terminology

Terpenes are assembled from C5 isoprenoid units. Monoterpenes are principally C10, sesquiterpenes C15, diterpenes C20, and triterpenes C30; oxygenated and rearranged products are more precisely terpenoids. Sesquiterpene lactones are C15 terpenoids containing a lactone ring and are generally treated as extractable specialized metabolites rather than ordinary essential-oil constituents.

Representative volatile classes include pinene, limonene, myrcene, camphor, 1,8-cineole, borneol, thujones, germacrene D, β-caryophyllene, α-humulene, davanone, spathulenol, and caryophyllene oxide. Important nonvolatile or less volatile chemistry includes artemisinin and related compounds, guaianolide and germacranolide sesquiterpene lactones, and triterpenoids associated with plant surfaces and defense.

Evidence map by class

Table 1. Major terpene classes, analytical contexts, and interpretation limits in *Artemisia*.

Class	Representative *Artemisia* chemistry	Main analytical context	Review interpretation
Monoterpenes and oxygenated monoterpenes	α-/β-pinene, limonene, myrcene, camphor, 1,8-cineole, borneol, thujones, artemisia ketone	GC-MS/GC-FID of volatile oils and headspace fractions	Often abundant, but highly sensitive to chemotype, tissue, stage, drying, and distillation
Sesquiterpene hydrocarbons and oxygenated sesquiterpenes	germacrene D, β-caryophyllene, α-humulene, davanone, spathulenol, caryophyllene oxide	GC-MS of oils; targeted volatile profiling	Useful markers of population chemistry, not universal species signatures
Sesquiterpene lactones	artemisinin, arteannuin B, artemisinic acid, santonin, guaianolides, germacranolides, absinthin-related compounds	LC-MS/HPLC, NMR, isolation; GC is often unsuitable without derivatization	Usually nonvolatile; structural and stereochemical confirmation is critical
Diterpenes	species-specific diterpenoid constituents and chlorophyll-associated phytol derivatives	Solvent extraction, LC-MS, NMR; often reported in broad phytochemical surveys	Less comprehensively sampled than C10/C15 chemistry; presence should be linked to isolation or validated MS evidence
Triterpenes and sterols	β-amyrin/α-amyrin-, friedelin-, cycloartane- and sterol-associated reports	Solvent extraction, LC-MS, GC-MS after derivatization, NMR	Surface and storage chemistry; not part of ordinary essential-oil comparisons

The unevenness of this evidence map is itself a result: volatile mono- and sesquiterpenes are overrepresented because they are accessible to routine GC analysis, whereas nonvolatile diterpenes, triterpenes, and lactones require extraction workflows that are less directly comparable across studies.

Primary studies show that the under-sampled classes are not merely theoretical. In *A. annua*, seed isolation recovered fourteen sesquiterpenes, three monoterpenes, and one diterpene (Brown et al. 2003). Separate trichome transcriptome and heterologous-expression work functionally characterized the oxidosqualene cyclase OSC2 and P450 CYP716A14v2 in aerial-cuticle triterpenoid production (Moses et al. 2015), while a diterpene-synthase study identified ten *A. annua* diTPS genes in a glandular-trichome and stress-resilience context (Chen et al. 2021). These are class-specific anchors, not evidence that all *Artemisia* species share the same diterpene or triterpene repertoire.

Monoterpenes and volatile sesquiterpenes

Monoterpenes are the most method-accessible *Artemisia* terpenoids. Across oils, recurrent chemistries include pinene and limonene hydrocarbons, camphor, borneol, 1,8-cineole, thujones, and irregular-monoterpene products such as artemisia ketone. Volatile sesquiterpenes add germacrene D, β-caryophyllene, α-humulene, caryophyllene oxide, davanone, spathulenol, and related oxygenated products. These names should be treated as reported identities with confidence levels, not interchangeable biomarkers: retention-index agreement, authentic-standard co-injection, mass-spectrum library matching, and NMR confirmation provide different evidentiary strength.

The biological unit is often a glandular-trichome or aerial-tissue oil, not a purified constituent. In *A. annua*, three recombinant monoterpene synthases generated distinct product spectra, while wounding and hormone treatment changed transcript levels. This connects volatile chemistry to enzyme specificity and inducible regulation, but it does not show that the induced oil is antiparasitic. In *A. frigida* and *A. absinthium*, population, geography, and phenology alter the relative abundance of volatile compounds. A cross-species comparison should therefore model presence/absence, relative abundance, and analytical detectability separately.

Enzyme studies reveal why one-gene/one-compound annotation is often inadequate. Recombinant *A. annua* epi-cedrol synthase produced epi-cedrol as its major oxygenated C15 product but also generated several olefinic sesquiterpenes and showed lower activity with the C10 precursor GPP (Mercke et al. 1999). Two additional *A. annua* sesquiterpene-cyclase cDNAs and later transient-expression experiments expanded the functionally tested synthase repertoire (Van Geldre et al. 2000; Kanagarajan et al. 2012). Product multiplicity means that an expanded TPS clade can increase chemical opportunity without predicting a unique metabolite, while heterologous expression can reveal catalytic potential that may be constrained by native substrate supply or cell type.

Downstream enzymes further diversify volatile products. A glandular-trichome-specific *A. annua* alcohol dehydrogenase oxidized monoterpene alcohol substrates in recombinant assays, demonstrating that volatile profiles reflect not only cyclization but subsequent redox chemistry (Polichuk et al. 2010). At the regulatory level, AabHLH112 perturbation changed β-caryophyllene, epi-cedrol, and β-farnesene-associated sesquiterpene metabolism, connecting a transcription factor to several non-artemisinin branches rather than a single endpoint (Xiang et al. 2022). Conversely, suppression of β-caryophyllene synthase increased artemisinin-pathway output in transgenic plants, providing direct evidence that FPP-consuming branches can compete for precursor under at least some engineered conditions (Chen et al. 2011).

Population studies show an orthogonal source of complexity. Oils of *A. campestris* and *A. herba-alba* varied among natural populations, and their non-parasitic bioassay profiles varied with composition (Younsi et al. 2017). In *A. herba-alba*, analysis of 80 individuals from eight Tunisian populations resolved four oil chemotypes and found substantial within-population marker diversity; global chemical divergence was not reducible to geographic or bioclimatic distance alone (Younsi et al. 2018). These observations argue for individual- or population-level sampling and against using a single oil chromatogram as a species diagnostic.

Diterpenes and triterpenes

Diterpene evidence is comparatively sparse and often mixed into broad phytochemical surveys. The *A. annua* seed study provides isolation-level evidence for one diterpene among three monoterpenes and fourteen sesquiterpenes, while a dedicated *A. annua* study identified ten diterpene synthases in relation to glandular trichome formation, artemisinin production, and stress resilience. A multi-platform *A. absinthium* oil analysis additionally reported diterpene- and homoditerpene-associated constituents alongside GC and NMR confirmation (Judzentiene et al. 2009). These findings support a real C20 biosynthetic dimension but not a genus-wide diterpene chemotype. Diterpene records should preserve seed versus leaf or trichome tissue, extraction polarity, and whether the compound was isolated or only assigned by MS.

Triterpenes and sterols are especially important for avoiding a false "essential-oil-only" view of *Artemisia*. OSC2 and CYP716A14v2 were functionally connected to α-, β-, and δ-amyrin-related cuticular triterpenoids in *A. annua*, and triterpenoids have also been isolated from *A. igniaria*. Their cuticle or surface-associated localization suggests roles in barrier and stress biology, but current records do not support transferring those roles to antiparasitic action. Because triterpenes are less volatile, a GC-only survey can systematically under-report them.

The current diterpene and triterpene literature remains too sparse and methodologically heterogeneous for a genus-wide abundance synthesis. A defensible survey should distinguish a structurally isolated C20 or C30 metabolite from a tentative LC-MS feature, and it should separate pathway-gene discovery from measured products. The *A. annua* diterpene-synthase study is unusually informative because it combined gene discovery with developmental and physiological perturbation, whereas the OSC2/CYP716A14v2 work paired trichome comparisons with recombinant functional tests. These studies provide reference loci for comparative genomics, but their products must be re-measured in homologous tissues before copy-number differences are interpreted as ecological or antiparasitic adaptations.

Sesquiterpene lactones

Sesquiterpene lactones (STLs) include guaianolide, germacranolide, eudesmanolide, and related structural families. Their α-methylene-γ-lactone or other electrophilic motifs can react with nucleophiles, which provides a chemically plausible basis for bioactivity but not a demonstrated parasite target. In *Artemisia*, STL reports include artemisinin-related amorphane chemistry, santonin, and numerous guaianolide and germacranolide constituents. Analytical workflows differ from oil profiling: LC-MS, isolation, multidimensional NMR, and stereochemical assignment are often needed, while direct GC percentages are not a suitable universal denominator.

The best-defined biosynthetic model for a non-artemisinin STL branch begins with FPP cyclization to germacrene A, oxidation to germacrene A acid, and downstream P450-dependent oxidation and lactonization. The presence of a germacranolide-like metabolite is therefore compatible with several levels of evidence-from structural analogy to a fully demonstrated enzyme sequence-and those levels must not be collapsed. STL abundance should also be reported separately from volatile sesquiterpene abundance because extraction and storage can partition them differently. In *A. annua*, field measurements of glandular-trichome dynamics across leaf development and senescence, together with dead-versus-green leaf precursor profiles, show why tissue age and precursor conversion must be retained as explicit variables (Lommen et al. 2006; Lommen et al. 2007).

The artemisinin branch demonstrates how metabolite isolation, enzyme biochemistry, and nonenzymatic chemistry can be integrated. Dihydroartemisinic acid and its hydroperoxide were isolated as candidate late intermediates before genetic perturbation established a broader branch architecture (Wallaart et al. 1999a; Wallaart et al. 1999b). CYP71AV1 disruption subsequently redirected metabolites toward arteannuin X and supported a major role for spontaneous oxidative chemistry in terminal artemisinin formation. The evidence therefore supports a hybrid pathway model: enzyme-controlled scaffold formation and oxidation establish the precursor pool, while storage environment and post-biosynthetic oxidation help determine terminal products.

Non-artemisinin lactones broaden both structural and pharmacological space. *A. gorgonum* yielded isolated guaianolide and germacranolide compounds with antiplasmodial testing, while *Artemisia*-derived or *Artemisia*-associated lactones such as dehydroleucodine have been tested against trypanosomatids. Dehydroleucodine inhibited cultured *T. cruzi* epimastigote growth and altered parasite physiology in a defined-compound study (Brengio et al. 2000). Such studies are stronger for constituent attribution than an oil assay, but species provenance, parasite stage, exposure, and host-cell selectivity still prevent direct translation to therapeutic efficacy.

Biosynthesis and compartmentalization

The plastidial MEP pathway and cytosolic mevalonate pathway supply IPP and DMAPP, which are condensed into prenyl diphosphates. GPPS and related enzymes provide C10 substrates for monoterpene synthases; FPPS provides the C15 substrate FPP for sesquiterpene synthases, including germacrene A synthase and ADS. Cytochrome P450s, reductases, dehydrogenases, oxidases, acyltransferases, glycosyltransferases, and spontaneous reactions expand the scaffold space.

Functionally characterized *A. annua* monoterpene synthases illustrate why copy number alone is insufficient: recombinant AaTPS2, AaTPS5, and AaTPS6 produced different product spectra, and transcript abundance changed after wounding and hormone treatments (Ruan et al. 2016). Product specificity, inducibility, and tissue localization therefore need to be modeled together in evolutionary comparisons.

The artemisinin pathway in *A. annua* is the strongest gene-to-metabolite reference. ADS converts FPP to amorpha-4,11-diene, a foundational enzyme result established in early biochemical work (Bouwmeester et al. 1999); CYP71AV1 oxidizes pathway intermediates; DBR2 and ALDH1 support the route toward dihydroartemisinic acid, after which terminal chemistry includes nonenzymatic steps. Glandular secretory trichomes provide a relevant biosynthetic and storage context. These statements are pathway anchors, not proof that homologous genes in another species produce artemisinin or confer antiparasitic activity.

The pathway was assembled through convergent evidence rather than a single experiment. ADS cloning and bacterial expression established the first committed cyclization; a trichome-enriched cDNA strategy identified CYP71AV1 and demonstrated sequential oxidation of amorpha-4,11-diene and related intermediates; and DBR2 cloning showed reduction at the branch that favors dihydroartemisinic-acid-derived chemistry (Chang et al. 2000; Teoh et al. 2006; Zhang et al. 2008). Stable-isotope feeding in intact growing plants supplied an independent carbon-flow perspective (Schramek et al. 2010). Together, these data are considerably stronger than pathway reconstruction from annotation alone, yet even this reference pathway remains sensitive to developmental state, precursor competition, and terminal nonenzymatic conversion.

Compartment evidence is equally important. Immunolocalization placed key artemisinin enzymes in apical cells of glandular secretory trichomes, and a trichome transcriptome enriched the candidate space for isoprenoid and downstream-modification genes (Olsson et al. 2009; Wang et al. 2009). Tissue-expression studies showed that terpene-metabolism genes are not uniformly expressed across roots, stems, leaves, and reproductive structures (Olofsson et al. 2011). A grafting study further implicated root-derived regulation of shoot trichomes and artemisinin-related metabolites, indicating that the producing cell is embedded in a whole-plant signaling system (Wang et al. 2016).

Precursor supply and branch competition have been tested genetically. Bidirectional perturbation of AaHDR1 altered artemisinin and other terpenoids, supporting a role for MEP-pathway supply while also showing class-specific responses rather than a uniform increase (Ma et al. 2017). Suppression of β-caryophyllene synthase shifted carbon away from one sesquiterpene branch, and a T-DNA insertion line produced a broader change in terpenoid composition and crude-extract bioactivity (Chen et al. 2011; Karaket et al. 2014). These perturbations support network competition, but engineered responses should not be assumed to describe standing natural variation.

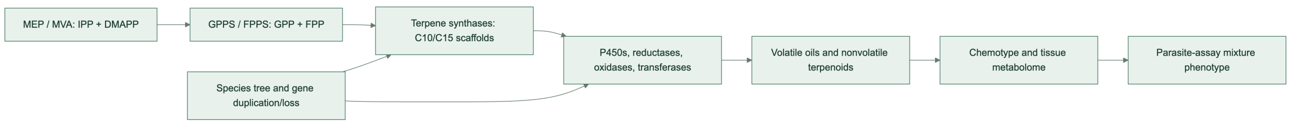
Regulation is multilayered. Jasmonate-responsive AaERF1/AaERF2, glandular-trichome-specific WRKY1, AaMYB108, and the AaGSW1-AaYABBY5/AaWRKY9 complex have each been connected to artemisinin-pathway expression or metabolite accumulation through transgenic, interaction, and expression experiments (Yu et al. 2012; Chen et al. 2017; Liu et al. 2023; Kayani et al. 2023). The emerging model is therefore not a simple linear pathway but a tissue-localized, environment-responsive network in which developmental programs and stress signals act on precursor supply, synthase branches, oxidation, and trichome capacity.

Germacranolide biosynthesis offers a second comparative framework: FPP can be cyclized by germacrene A synthase, oxidized by germacrene A oxidase, and converted through P450-dependent chemistry and lactonization. The broad presence of a scaffold family does not imply identical product profiles because enzyme specificity, tissue localization, substrate competition, and downstream modification differ.

In *A. annua*, a glandular-trichome cDNA library yielded a germacrene A synthase that converted FPP to germacrene A in heterologous assays (Bertea et al. 2006). This creates an experimentally anchored comparator for germacranolide-like pathways and illustrates a general strategy for other species: first establish synthase product, then test downstream oxidases, tissue expression, and matching metabolites. Sequence similarity to germacrene synthase, by itself, is insufficient because closely related TPS proteins can differ in product spectra and because downstream enzymes may redirect a shared scaffold.

Pathway evidence boundary. The ordered pathway model is a set of evidence-qualified hypotheses. Direct enzyme characterization supports individual steps; a homologous sequence or transcript supports candidate function; a detected metabolite supports presence in the sampled material. None of these alone establishes flux, compartment, or in-vivo production in a second species. Multi-substrate requirements, physiological direction, stereochemistry, and missing transport or cofactor information must remain explicit in the Terpedia pathway records.

Figure 1. Conceptual route from precursor supply to phenotype.



Species and chemotype diversity

A. annua combines artemisinin-pathway chemistry with variable volatile oils that may include artemisia ketone, camphor, germacrene D, and 1,8-cineole. *A. absinthium* can show thujone-, camphor-, cineole-, davanone-, or chamazulene-associated profiles depending on provenance and stage. *A. herba-alba*, *A. vulgaris*, and *A. argyi* likewise show regional and tissue-level variation. Multi-species Canadian oils reinforce that composition can differ substantially among taxa and regions (Lopes-Lutz et al. 2008). *A. dracunculus* is a useful boundary case because estragole may dominate some oils; estragole is not a terpene and should remain in a broader volatile-metabolite record.

Geography and climate can alter both the relative abundance and the dominant chemical class. Phenological shifts may change monoterpene and sesquiterpene proportions between vegetative and flowering stages. Drying and distillation can remove, transform, or concentrate constituents. These effects explain why percentage tables cannot be pooled without specimen-level metadata and analytical harmonization.

Early seasonal studies of diploid and tetraploid *A. annua* and accessions from different geographic origins demonstrated that artemisinin, artemisinic acid, dihydroartemisinic acid, and related precursors do not covary uniformly (Wallaart et al. 1999; Wallaart et al. 2000). A later multi-germplasm field study likewise found distinct seasonal trajectories among Brazilian, Chinese, and Swiss materials and argued that selection on both artemisinin and dihydroartemisinic acid could be more informative than selection on artemisinin alone (Ferreira et al. 2018). These studies turn “chemotype” from a label into a multivariate pathway phenotype.

Genetic and environmental components can be partially separated but rarely ignored. A linkage map identified loci affecting artemisinin yield in *A. annua*, providing direct evidence that segregating genetic variation contributes to the phenotype (Graham et al. 2010). In contrast, salt exposure, phytohormone treatment, reproductive development, and methyl jasmonate each changed pathway or oil phenotypes within experimental genotypes (Yadav et al. 2017; Maes et al. 2011; Arsenault et al. 2010; Wu et al. 2011). The DBR2-promoter study further linked promoter activity to high- and low-artemisinin chemotypes, showing how cis-regulatory variation at a branch point can contribute to pathway allocation (Yang et al. 2015).

The same principle applies to apparently stable markers. Germacrene D is dominant in many *A. frigida* samples, yet the distribution of that dominance and its relationship to climate must be evaluated across populations and years. Recent *A. absinthium* work similarly shows that geography and phenological stage can change the dominant oxygenated mono- versus sesquiterpene class. These studies support stratified comparisons rather than a single canonical “wormwood oil.”

Post-harvest handling is part of the phenotype that reaches an assay. Controlled drying changed concentrations of artemisinin and its acid precursors in *A. annua* leaves (Ferreira and Luthria 2010). Hydrodistillation, solvent extraction, headspace analysis, and direct tissue extraction sample different chemical spaces and may also create or remove products through heat, oxidation, hydrolysis, or volatility. A reproducible comparison must therefore treat extraction method, storage duration, temperature, moisture endpoint, and abundance denominator as model variables rather than descriptive footnotes.

Developmental series in *A. annua* make the confounding concrete: artemisinin and related sesquiterpenes were highest at particular vegetative stages, while the essential-oil mono:sesquiterpene ratio shifted from approximately 55:45 near the preflowering maximum to approximately 30:70 at other stages (Woerdenbag et al. 1994). A separate two-genotype time series identified genotype- and stage-dependent marker sets that included artemisinic acid, arteannuin B, borneol, beta-farnesene, camphor, and lanceol (Wang et al. 2009). Thus, “chemotype” should be modeled jointly with development and genotype rather than treated as a static label.

A five-species comparison provides a useful cross-species test of this principle: variation in glandular-trichome density, artemisinin content, and expression of pathway genes was observed among *A. annua*, *A. deserti*, *A. absinthium*, *A. sieberi*, and *A. khorassanica* (Salehi et al. 2018). This is stronger comparative evidence than extrapolating from a single *A. annua* genotype, but expression and metabolite association still require common-annotation orthology, enzyme assays, and matched specimen chemistry before they can be interpreted as evolutionary gain or loss.

Artemisia argyi as a tissue- and time-resolved comparator

The expanding *A. argyi* literature provides a useful counterweight to *A. annua*. De novo and full-length transcriptomes identified candidate genes across the MVA/MEP backbones, TPS families, P450s, and regulatory families, but these resources differ in tissue composition, assembly strategy, and annotation granularity (Liu et al. 2018; Cui et al. 2021). Targeted terpenoid measurements paired with bHLH-family analysis nominated regulatory candidates without claiming direct enzyme function (Yi et al. 2021).

Tissue-resolved metabolomics and transcriptomics showed that roots, stems, and leaves differ in both terpene-associated metabolites and candidate pathway expression; leaf-enriched MEP-pathway genes and co-expressed P450s formed hypotheses for subsequent functional testing (Zhang et al. 2022). Temporal profiling across four harvest months reported changing volatile-oil levels and identified full-length transcripts correlated with monoterpene and sesquiterpene profiles (Xu et al. 2022). These studies demonstrate that tissue and harvest date can produce differences as large as those attributed to genotype.

Single-cell transcriptomics and matched metabolomics now resolve this problem at higher spatial resolution. Glandular trichomes and non-glandular tissues of *A. argyi* showed strongly different metabolite profiles, with sesquiterpenoids prominent among trichome-enriched terpenoid differences, and the single-cell atlas nominated developmental and TPS-expression programs (Dong et al. 2026). This is powerful candidate evidence, but co-expression edges and cell-type enrichment remain hypotheses until recombinant enzymes, gene perturbations, and isotope or product-feeding experiments establish catalytic sequence and flux.

Comparative evolutionary genomics

The initial controlled comparison is *A. annua* versus *A. argyi*. The former supplies a pathway-rich reference genome and functional artemisinin literature. The latter has a chromosome-scale genome with reported whole-genome duplication and terpene-synthase expansion, plus tissue transcriptomes. The apparent ADS contradiction is now interpretable as an assembly-, ploidy-, annotation-, and function-level discrepancy rather than a simple presence/absence result. The earlier chromosome-scale study selected a 3.89-Gb, 17-pseudochromosome haplotype-A projection, annotated 122 TPS genes, and described partial ADS deletion and loss of function. A later 7.88-Gb haplotype-resolved assembly predicted 451 TPS models, of which 261 were manually classified as intact, and identified a six-copy tandem ADS homolog cluster; recombinant AarADS proteins produced α-bisabolol rather than the amorpha-4,11-diene made by AanADS (Chen et al. 2023; Qi et al. 2025). Thus, the later result supports ADS-family retention with functional divergence, while neither paper alone establishes an intact, expressed, artemisinin-producing pathway in *A. argyi*.

The comparative analysis should infer orthogroups and gene trees for GPPS, FPPS, monoterpene synthases, sesquiterpene synthases, ADS, CYP71AV1-like P450s, DBR2, ALDH1, OSCs, and selected sesquiterpene-lactone enzymes. Duplications and losses should be mapped to a broad *Artemisia* species tree. Expression should be interpreted by tissue and developmental context, with glandular trichome data prioritized where available. Missing annotation is unresolved data, not evidence of biological absence.

Published comparative results already provide testable anchors: the later *A. argyi* analysis reports 451 predicted TPS models split between two haplotypes (221 and 230), with 261 intact models after manual correction, while the chromosome-scale study links WGD and duplication modes to TPS expansion and identifies a germacrene D synthase cluster. The latter study also reports pathway elucidation for germacrenes, (+)-borneol, and (+)-camphor. These observations are catalogued in the KB's `gene-family-evidence.csv` and ADS/TPS reconciliation table; they are not treated as a substitute for a common-annotation, orthology-aware rerun.

The transcript and metabolite literature provides a validation ladder for genomic predictions. A candidate duplicated TPS is more compelling when it is intact under a common annotation, placed within a supported gene tree, retained in synteny or a tandem cluster, expressed in the relevant secretory cell, and correlated with a product that the recombinant protein can generate. The *A. argyi* tissue, temporal, and single-cell studies supply several of these intermediate layers, but none independently establishes orthology or ancestral function. In *A. annua*, by contrast, the mature ADS/CYP71AV1/DBR2 evidence chain shows what eventual convergence should look like.

Regulatory evolution should be evaluated with the same caution as enzyme evolution. The abundance of MYB, bHLH, WRKY, AP2/ERF, and light- or jasmonate-responsive candidates in transcriptomic analyses may reflect both genuine network expansion and permissive co-expression thresholds. Functional studies of AaERF1/AaERF2, WRKY1, AaMYB108, AabHLH112, and the AaGSW1 complex define experimentally testable motifs and interactions in *A. annua*, but homologous regulators in *A. argyi* should not inherit those functions by name alone. Conserved target motifs, protein interactions, cell-type expression, perturbation phenotypes, and metabolite responses are required to claim network conservation.

The Terpedia KB now carries an execution specification for this analysis (`comparative-genomics/`). Its required outputs include frozen input checksums, software versions, orthology-aware gene trees, synteny/tandem-cluster calls, and a tiered gene-family evidence table. Until those outputs are generated and reviewed, the genome comparison in this article is a design-supported hypothesis framework rather than a completed comparative result.

An NCBI Datasets metadata audit was run without downloading sequence archives. It returned one *A. annua* assembly report and three separate *A. argyi* reports. The literature resource `PRJCA010808` was resolved to its external NGDC/CNCB project record, not to an NCBI assembly accession; the published 3.89-Gb, 17-pseudochromosome haplotype must therefore be treated as a distinct input until sequence and annotation files are obtained and checksummed. This is itself a result for reproducibility: assembly choice must be documented before copy-number or presence/absence comparisons are interpreted.

The KB's assembly comparison table makes the confounders explicit: the *A. annua* NCBI reference is scaffold-level, the literature *A. argyi* input is a 3.89-Gb haplotype-A projection, and NCBI also exposes larger or differently assembled *A. argyi* resources. Genome length and chromosome count should therefore not be interpreted as evidence for terpene-family expansion without a common annotation and orthology workflow.

The execution gate is also explicit: NCBI reports an annotation with 63,226 protein-coding genes for the *A. annua* reference, and the earlier NGDC/CNCB *A. argyi* haplotype-A annotation exposes 62,844 protein-coding genes. These counts are not directly comparable to the later 7.88-Gb haplotype-resolved annotation. A seed-centered DIAMOND screen of 31 annotated *A. annua* pathway/TPS queries returned 105 *A. argyi* candidate hits; the exact inputs and hashes are archived in the KB. These are homology candidates only. A cross-species copy-number or orthology result is not reported until a common annotation, reciprocal/phylogenetic validation, and gene-tree reconciliation are completed.

Reciprocal checking re-searched 67 unique *A. argyi* candidates against the *A. annua* proteome and produced 26 reciprocal-best-hit candidate rows. Because a duplicated *A. argyi* locus can share the same best *A. annua* seed, this count is not a one-to-one ortholog count and is not interpreted as gene-family expansion.

As a diagnostic next step, the KB built MAFFT alignments and IQ-TREE3 exploratory trees for the three reciprocal groups that retained at least four unique protein sequences (PWA60940.1, PWA68009.1, and PWA85324.1). The run used the LG+G4 model, 1,000 ultrafast-bootstrap replicates, 1,000 SH-aLRT replicates, and one thread; the exact alignments, tree files, reports, software versions, and input checksums are archived under `comparative-genomics/results/seed-gene-trees-2026-09-01/`. Eight of eleven reciprocal groups were not tree-built because they had fewer than four leaves or fewer than four unique sequences. These small, seed-biased trees are quality-control diagnostics only: they do not establish orthology, duplication/loss, or pathway function.

A restricted OrthoFinder diagnostic then clustered the 31 *A. annua* seed proteins and 67 unique *A. argyi* forward-hit candidates. It produced 16 candidate orthogroups, 11 shared clusters, and five *A. argyi*-only clusters. The full tables, input FASTAs, software context, and checksums are archived in `comparative-genomics/results/candidate-orthofinder-2026-09-01/`. This result is useful for prioritizing follow-up gene trees and synteny, but the seed-biased, two-taxon candidate set cannot support genome-wide orthology, copy-number, duplication/loss, or gene-function claims.

An independent *A. argyi* leaf, root, and stem transcriptome supplies candidate terpenoid-biosynthesis transcripts and tissue contrasts (Liu et al. 2018). It extends the comparison beyond *A. annua*, but de novo transcript annotation cannot distinguish paralogs, homeologs, and allelic copies reliably enough for gene-family evolution without genome-guided placement. Experimental work on an *A. annua* (E)-β-farnesene synthase further shows that a sparse set of epistatic sequence changes can alter terpene-enzyme behavior (Ballal et al. 2020); this motivates testing sequence-function effects within orthologous clades rather than treating all TPS copies as functionally equivalent.

Environmental and perturbation studies provide useful mechanistic tests within *A. annua*. Short-term UV-B exposure increased artemisinin production and induced ADS, CYP71AV1, and stress/hormone-associated transcripts (Pan et al. 2014). A CYP71AV1 mutant redirected flux toward the alternate sesquiterpene epoxide arteannuin X and supported nonenzymatic terminal conversion in glandular trichomes (Czechowski et al. 2016). These results demonstrate inducibility and metabolic plasticity, but they do not establish that a homologous response or alternate product occurs in another *Artemisia* species.

Antiparasitic evidence

Two parasite-facing stories deserve priority in this review. Malaria is the strongest mechanistic case because artemisinin has a chemically defined scaffold, a reconstructed *A. annua* pathway, extensive parasite pharmacology, and a clinically established artemisinin-based combination-therapy context. Wormwood (*A. absinthium*) is a separate, important veterinary anthelmintic case: traditional vermifuge use is supported by several extract, oil, feed, and sheep studies, but the active chemistry, dose standardization, and human safety remain unresolved. The evidence should be developed in parallel rather than merged under the generic label "antiparasitic *Artemisia*."

Terpene-protein interaction map

The new Terpedia interaction map separates target engagement from target hypotheses (`parasite-protein-interactions.csv`; extraction rules in `protein-interaction-protocol.md`). The most experimentally anchored case is artemisinin against *Plasmodium falciparum*: haem activation supports covalent labeling of a broad parasite protein set, with 124 reported targets and orthogonal validation of selected metabolic and structural proteins, including OAT, PyrK, LDH, SpdSyn, SAMS, plasmepsins, MSP1, and actin (Wang et al. 2015). This is a promiscuous, chemically activated interaction model, not evidence for one universal artemisinin receptor.

Additional malaria studies provide mechanistic anchors at different evidence levels. DHA inhibited immunopurified PfPI3K at nanomolar concentration, and PfK13 resistance biology was linked to altered PfPI3K/PI3P regulation; a mammalian-kinase counter-screen supported parasite selectivity in that experiment (Mbengue et al. 2015). Artemisinin binding to recombinant PfTCTP was mapped by surface plasmon resonance, mass spectrometry, and structural analysis, although the contribution of this interaction to parasite killing remains unresolved (Eichhorn et al. 2013). Functional oocyte and parasite-imaging experiments supported iron-dependent PfATP6 inhibition, but later multi-target evidence argues against treating PfATP6 as the sole receptor (Eckstein-Ludwig et al. 2003). Separately, DHA caused protein damage and compromised parasite proteasome function, producing proteostasis stress (Bridgford et al. 2018). A stage-resolved photoaffinity study reported covalent and noncovalent target classes associated with protein synthesis, glycolysis, and oxidative homeostasis (Gao et al. 2024); cryo-EM and MS-CETSA subsequently showed that artemisinin treatment increases PfRACK1 occupancy on Pf80S ribosomes (Go et al. 2025). Together these records favor a multi-target, stage- and activation-dependent model; they do not make PfPI3K, the proteasome, PfTCTP, PfATP6, or PfRACK1 a sole mechanism.

Evidence is weaker outside malaria. A 123-compound sesquiterpene docking study proposed *Leishmania* PTR1, carbonic anhydrase, NMT, and trypanothione synthetase as binding hypotheses; xanthanolides 75-76 had the strongest reported predicted PTR1 scores (-10.6 kcal/mol), but this is E2 docking evidence and the compounds are not established as *Artemisia* constituents in that record (Mishra et al. 2014). In *T. brucei*, recombinant-enzyme assays strengthened the case for target-class activity: cnicin inhibited TbPTR1, while cynaropicrin inhibited both TbPTR1 and TbDHFR, with IC50 values of 12.4 and 7.1 μM, respectively; both compounds came from non-*Artemisia* Asteraceae sources (Possart et al. 2021). A cruzain docking paper is retained as *T. cruzi* context only, not as a demonstrated *Artemisia* interaction. For *H. contortus*, the current *Artemisia* records establish anthelmintic phenotypes but no parasite protein target, making target discovery a high-priority gap rather than an inferred mechanism.

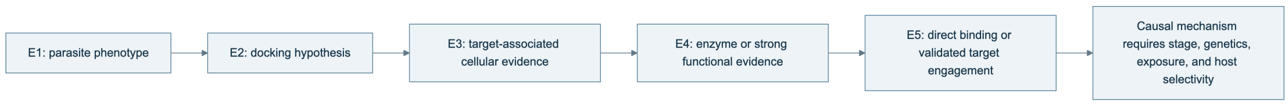
The practical conclusion is that artemisinin supplies a validated parasite-protein interaction framework, while most non-artemisinin *Artemisia* terpene claims still need purified-compound assays, parasite permeability measurements, target-rescue or genetic tests, and host-protein counterscreens. The KB therefore stores interaction mode, evidence tier, compound origin, parasite stage, and inference boundary for every record.

Table 2. Cross-parasite terpene-protein evidence landscape.

Parasite	Compound context	Protein or complex	Strongest current evidence	Boundary
P. falciparum	Artemisinin/DHA	Broad covalent target set	E5 activity-based proteomics with selected orthogonal validation	Probe- and activation-dependent; not all targets are equally causal
P. falciparum	Artemisinin/DHA	PfPI3K, PfTCTP, PfATP6, proteasome, PfRACK1-Pf80S	E3-E5 biochemical, functional, genetic, structural, and complex-level evidence	Supports a multi-target model, not a single universal receptor
Leishmania spp.	Sesquiterpene-related panel	PTR1, carbonic anhydrase, NMT, TryS	E2 docking	Requires direct binding, enzyme inhibition, target engagement, and *Artemisia* provenance
T. brucei	Cnicin and cynaropicrin	TbPTR1 and TbDHFR	E4 recombinant-enzyme inhibition plus docking	Compounds were sourced from non-*Artemisia* Asteraceae; parasite target engagement remains open

Parasite	Compound context	Protein or complex	Strongest current evidence	Boundary
T. cruzi	Terpenoid panel	Cruzain	E2 docking with prior inhibition context	Not an *Artemisia*-matched direct interaction experiment
H. contortus	*Artemisia* oils, extracts, and one isolated sesquiterpene	Unresolved	E1 phenotype	Target discovery and host-selectivity experiments are required

Figure 2. Evidence ladder used for terpene-parasite target claims.



The arrows indicate increasing evidentiary requirements, not an automatic promotion path. A docking result advances only when the same compound-protein pair is tested experimentally, and even direct binding does not establish that the interaction is necessary or sufficient for parasite killing.

Published essential-oil studies provide useful phenotype anchors. *A. annua* leaf oil has been tested against *Leishmania donovani* in in-vitro and in-vivo contexts. Oils from *A. campestris* and *A. herba-alba* have been tested against *L. infantum* promastigotes, with apoptosis-like and cell-cycle-associated effects reported. These results support mixture-level activity under the reported experimental conditions. They do not establish a single active terpene, whole-life-cycle efficacy, host safety, or clinical benefit.

Whole-leaf and aqueous-extract studies occupy a different evidentiary category from essential oils. *A. annua* leaf powder was evaluated against *Leishmania* in cell and animal models, but the complex material prevents attribution to artemisinin or another terpene (Mesa et al. 2017). *A. nilagirica* extracts were screened against *P. falciparum*, again providing a species- and preparation-specific phenotype rather than a defined terpene mechanism (Panda et al. 2018). Most recently, aqueous *A. afra* and *A. annua* extracts were compared against artemisinin-resistant *P. falciparum* with selectivity measurements (Roesch et al. 2025). This resistant-parasite design is valuable, but activity of a chemically complex infusion cannot be projected onto an isolated constituent or clinical regimen.

The quantitative assay records are separated in `antiparasitic-evidence.csv` so that parasite stage, preparation, endpoint, dose, host control, and translation boundary remain visible. For example, the camphor-rich *A. annua* oil reported IC50 values of 14.63 +/- 1.49 microgram/mL against promastigotes and 7.3 +/- 1.85 microgram/mL against intracellular amastigotes, plus an approximately 90% liver/spleen burden reduction at 200 mg/kg in infected mice. Those observations belong to one hydrodistilled oil and its experimental model; they are not evidence that camphor alone, or *A. annua* generally, produces the effect.

An Ethiopian study of *A. abyssinica* further illustrates why selectivity must be reported with activity: its oxygenated-monoterpene-rich oil showed activity against *Leishmania* promastigote and amastigote systems but also variable toxicity in mammalian-cell and erythrocyte assays. This is a valuable comparative record because it carries parasite stage and host-toxicity information together, although it still does not resolve which constituents or interactions drive the result.

The broader antiparasitic record is heterogeneous but not limited to Leishmania. *A. campestris* oil has been tested against *Haemonchus contortus* eggs and adults and against *Heligmosomoides polygyrus* in a nematode model (Abidi et al. 2018); *A. vestita* and *A. maritima* extracts have been tested against *H. contortus* in vitro and in infected sheep (PubMed). A hydrodistilled *A. gilvescens* oil and selected constituents have also shown larvicidal activity against *Anopheles anthropophagus* (PubMed). Seven geographic *A. argyi* oils provide a population-level vector comparison against *Anopheles sinensis*, but the shared and variable constituents remain mixture-level predictors rather than identified actives (Luo et al. 2022). These studies widen the phenotype space to helminths and vectors, but they also emphasize that extract polarity, dose, life stage, host model, and formulation prevent direct pooling. An Ethiopian extract comparison against bloodstream *Trypanosoma brucei* adds another protozoan anchor while demonstrating why cytotoxicity and extract composition must accompany activity (PubMed).

Within helminth research, defined-compound and whole-preparation studies can be compared without treating them as equivalent. Artemisinin and *Artemisia* extracts were tested against *H. contortus* in a gerbil model, directly exposing the distinction between purified compound and botanical matrix (Squires et al. 2011). *A. lancea* essential oil was tested against egg and larval stages in vitro (Zhu et al. 2013), whereas bio-guided *A. cina* isolation later produced a defined sesquiterpene lead for L3 larvae. These assays suggest tractable phenotypes for fractionation and target discovery, but host pharmacokinetics, adult-worm activity, and safety remain separate questions.

Historical structure-activity work on artemisinin and related sesquiterpenes in rodent malaria already showed that close chemical relatives can differ substantially in activity (Peters et al. 1986). That observation remains relevant to comparative genomics: detecting a pathway-related scaffold or ADS-like sequence does not establish an artemisinin-equivalent phenotype. Structure, peroxide status, stereochemistry, exposure, and activation environment all matter.

Two additional high-priority records sharpen the chemistry-phenotype boundary. In *A. indica*, a hydrodistilled oil contained 92 detected compounds, with camphor (13.0%) and caryophyllene oxide (10.87%) as major constituents; the oil and solvent extracts showed activity across *Plasmodium falciparum*, *Trypanosoma brucei*, and *T. cruzi* systems, with several preparations showing low mammalian cytotoxicity (Tasdemir et al. 2015). The study is valuable because chemistry, parasite panels, and cytotoxicity were measured together, but it remains a mixture/extract study. In *A. scoparia*, a BA plus 2,4-D callus extract inhibited *Leishmania tropica* promastigotes with an IC50 of 19.13 mg/mL and produced apoptotic morphology; GC-MS reported nerolidol alongside non-terpene metabolites (Yousaf et al. 2019). This is evidence for a tissue-culture-derived crude extract, not proof that nerolidol or any single terpene accounts for the phenotype.

The strongest causal bridge would require matched specimen chemistry, parasite-stage assays, host-cell cytotoxicity, fractionation or defined mixtures, and ideally purified-compound plus combination testing. Gene expression or a predicted pathway can prioritize candidates but cannot replace those experiments.

Additional species and preparation types reinforce this boundary. *A. pedemontana* subsp. *assoana* essential oils and hydrolate were evaluated across insect, nematode, protozoan, and antiplasmodial assays, making it a useful multi-phenotype specimen record while retaining oil and hydrolate as distinct preparations (Sainz et al. 2019). Isolated *A. sieversiana* sesquiterpenoids provide a compound-level trypanocidal lead set (Nurbek et al. 2020), whereas *A. absinthium* fractionation studies link selected oil fractions to *T. cruzi* and *T. vaginalis* activity with cytotoxicity measurements (Martínez-Díaz et al. 2015). These records increase taxonomic and mechanistic coverage, but they do not make mixture, purified-compound, or host-selectivity endpoints interchangeable. A bio-guided *A. cina* study adds a stronger constituent-to-phenotype bridge by isolating a sesquiterpene active against *Haemonchus contortus* L3 larvae (Arango-De la Pava et al. 2024); an *A. brevifolia* study extends the nematode comparison across life stages (Hussain et al. 2025). Both still require host-selectivity and in-vivo confirmation.

Purified-compound evidence provides an important contrast to oil studies: helenalin, mexicanin, and dehydroleucodine inhibited cultured *Leishmania mexicana* promastigotes at approximately low-micromolar concentrations, but that result cannot be assumed for an *Artemisia* oil or a different parasite stage (Barrera et al. 2008). Likewise, methanolic aerial-part extracts of *A. sieberi* and *A. khorassanica* have been evaluated in *Plasmodium berghei* mouse models (PubMed; PubMed). These records broaden the evidence base while reinforcing the need to retain extraction polarity, dose, parasite model, and compound identity.

Isolation and fractionation can provide a stronger causal bridge than crude-oil screening. Leaves and flowers of Cape Verdean *A. gorgonum* yielded guaianolides and germacranolides; ridentin had a *P. falciparum* FcB1 IC50 of 3.8 +/- 0.7 microgram/mL against a Vero-cell IC50 of 71.0 +/- 3.9 microgram/mL (Ortet et al. 2008). Conversely, a Cuban *A. absinthium* oil dominated by trans-sabinyl acetate inhibited *L. amazonensis* promastigotes and amastigotes and reduced lesion and parasite burden after intralesional dosing in mice (Monzote et al. 2014). The contrast is informative: purified-lactone selectivity and oil-level in-vivo activity are both stronger than an uncharacterized extract claim, but neither supports extrapolation to all *Artemisia* preparations. A separate *A. absinthium* oil study tested the oil and three major constituents against six mosquito vectors while measuring acute toxicity in non-target aquatic organisms (Govindarajan and Benelli 2016); this is valuable paired efficacy-ecotoxicity evidence, but still preparation- and dose-specific.

Phylogeny can improve discovery prioritization without replacing chemical validation. Sampling of 117 ethnobotanically reported species detected artemisinin in nine species, including eight reported for the first time, but detection does not establish abundance, pathway completeness, or therapeutic equivalence (Pellicer et al. 2018). This finding supports a comparative design in which phylogenetic position selects accessions for targeted LC-MS and pathway-genomics follow-up rather than serving as a proxy for efficacy.

Malaria: artemisinin as the primary validated case

Malaria should be the anchor phenotype for the evolutionary-genomics component because it links a defined sesquiterpene lactone to a validated plant pathway and a clinically important parasite. The artemisinin scaffold contains a peroxide bridge whose activation is central to parasite damage, and the *A. annua* ADS-CYP71AV1-DBR2-ALDH1 network provides the best-resolved plant-side genotype-to-metabolite framework in the genus. Historical structure-activity work showed that related sesquiterpenes are not functionally interchangeable, while modern target studies support activation-dependent, multi-target parasite injury rather than one universal receptor (Peters et al. 1986; Wang et al. 2015).

The clinical boundary must remain explicit. WHO guidance supports quality-assured artemisinin-based combination therapy for malaria; it does not validate arbitrary teas, capsules, essential oils, or unstandardized *Artemisia* materials. Whole-plant experiments remain scientifically valuable for understanding matrix effects and resistance evolution. In rodent models, dried whole-plant *A. annua* overcame selected artemisinin resistance and delayed resistance acquisition relative to pure artemisinin (Elfawal et al. 2015). That result motivates fractionation, exposure, and resistance-mechanism studies; it does not establish human efficacy, dosing, quality control, or safety of a nonpharmaceutical whole-plant product.

For the comparative study, malaria-related claims should be separated into three levels: (1) *A. annua* artemisinin production and parasite mechanism, which have the strongest evidence; (2) detection of artemisinin or related lactones in another *Artemisia* species, which supports chemical occurrence only; and (3) an *Artemisia* extract phenotype against *Plasmodium*, which supports a preparation-level assay result. A species can satisfy level 2 or 3 without possessing the same pathway flux, peroxide abundance, pharmacokinetics, or resistance profile as *A. annua*. This tiering is central to the proposed orthology and metabolomics workflow.

Whole-leaf mutant experiments refine the frequently proposed "artemisinin plus flavonoid synergy" explanation. A *CHI1* mutant that lacked major flavonoids but retained wild-type artemisinin levels showed antiplasmodial activity comparable to wild type, whereas *CYP71AV1*- and ADS-impaired lines with severely reduced artemisinin biosynthesis showed little or no activity against intraerythrocytic *P. falciparum* Dd2 (Czechowski et al. 2019). In that matched genetic system, flavonoids were not a major independent driver of rapid in-vitro activity. This does not eliminate matrix effects in other cultivars, extraction methods, parasite strains, or exposure windows; it establishes a stronger causal comparator than correlation of unrelated metabolites.

Environment can also move both chemistry and apparent antimalarial potency. Under controlled LED treatments, white and blue light increased artemisinin and artemisinic acid relative to red light and shifted mono-, sesqui-, di-, and triterpene profiles. Crude-leaf extract IC50 values against *P. falciparum* NF54 were 0.51 and 0.52 microgram/mL under white and blue light, 1.11 microgram/mL under red light, and 2.23 microgram/mL for greenhouse-grown plants (Sankhuan et al. 2022). Because the experiment changed multiple metabolites together, the result supports an environment-chemotype-phenotype relationship rather than a causal assignment to germacrene D, trans-beta-farnesene, artemisinin, or any other correlated compound.

Wormwood (A. absinthium): vermifuge evidence and limits

Wormwood has a long-standing traditional reputation as a vermifuge, but the scientifically appropriate term for the current evidence is preparation-specific anthelmintic activity. The strongest early veterinary study compared crude aqueous and ethanolic aerial-part extracts with albendazole against ovine gastrointestinal nematodes. Both extracts impaired adult *Haemonchus contortus* motility in vitro and reduced fecal egg output in sheep; the ethanolic extract reached 90.46% fecal egg-count reduction at 2.0 g/kg body weight on day 15, whereas the aqueous extract was less active (Tariq et al. 2009). This is meaningful in-vivo veterinary evidence, but it does not identify a terpene, define a standardized extract, or establish use in humans.

Essential-oil evidence adds a different chemistry and exposure model. Oils from two cultivated, thujone-free *A. absinthium* populations were dominated by (Z)-epoxyocimene and chrysanthenol, varied quantitatively and qualitatively between populations, and reduced *Trichinella spiralis* infectivity ex vivo at 0.5-1 mg/mL without mammalian-cell cytotoxicity in that assay. The same study reported a 66% reduction of intestinal adults in vivo, while the oils were ineffective against the plant-parasitic nematode *Meloidogyne javanica* (García-Rodríguez et al. 2015). The population dependence and parasite selectivity are important: they argue for chemotype-by-parasite testing, not for a universal "wormwood oil" mechanism.

Feed and field studies are more heterogeneous. Dried wormwood supplementation and aqueous extracts produced strong in-vitro egg-hatch effects against *H. contortus*, with ED50 and ED99 values of 1.40 and 3.76 mg/mL for the wormwood extract, but fecal egg counts did not differ significantly at the group level in experimentally infected lambs (Mravčáková et al. 2020). A low-input swine-farm study reported antiparasitic effects after *A. absinthium* powder administration, but the co-occurring plant intervention and complex powder chemistry prevent attribution to terpenes (Băieş et al. 2024). A recent Awassi-sheep study of crude ethanolic extract reported 76.77% fecal egg-count reduction on day 15 compared with 100% for albendazole and no observed adverse effects during the small trial (Irehan et al. 2026). The latter is useful contemporary evidence, but it remains a single regional study with 18 animals and requires independent chemical, residue, dose, and safety confirmation.

Wormwood also has a defined-compound trematode signal that should not be conflated with the ovine nematode results. A dichloromethane *A. absinthium* extract caused 100% mortality of adult *Schistosoma mansoni* at 200 microgram/mL in vitro, and fractionation yielded the flavone artemetin and the sesquiterpene lactone hydroxypelenolide (Almeida et al. 2016). The study supports a preparation-level schistosomicidal phenotype and identifies chemically tractable leads, but it does not establish the isolated compounds' potency, selectivity, pharmacokinetics, or efficacy in an infected host.

Formulation can change the exposure boundary independently of plant chemistry. A Cuban *A. absinthium* essential oil formulated in nanocochleates showed an intracellular-*L. amazonensis* IC50 of 21.5 +/- 2.5 microgram/mL against a macrophage IC50 of 27.7 +/- 5.6 microgram/mL and suppressed murine cutaneous infection by approximately 50% after intralesional dosing at 30 mg/kg (Tamargo et al. 2017). This is a useful formulation and host-counterassay record, not evidence that unformulated wormwood oil is a human vermifuge or that a particular terpene is responsible.

Table 3. Focused malaria and wormwood evidence anchors. Values are retained at the preparation, parasite-stage, and host-model level; they are not pooled effect estimates.

Case	Preparation and observed signal	Evidence boundary
A. annua / *P. falciparum* Dd2	Matched whole-leaf mutant extracts: a flavonoid-deficient *CHI1* mutant retained wild-type artemisinin and antiplasmodial activity; artemisinin-impaired lines showed little or no activity	Supports artemisinin dominance in this genetic system; matrix effects elsewhere remain unresolved
A. annua / *P. falciparum* NF54	LED-conditioned leaf extracts: IC50 0.51, 0.52, 1.11, and 2.23 microgram/mL for white, blue, red, and greenhouse plants	Environment-chemotype-phenotype association; no single terpene assigned as causal
A. annua / rodent malaria	Dried whole plant versus artemisinin: whole plant overcame selected *P. yoelii* resistance and slowed stable *P. chabaudi* resistance	Preclinical matrix result; does not replace quality-assured ACT or establish human efficacy
A. absinthium / ovine gastrointestinal nematodes	Crude aerial-part extracts: ethanolic extract reached 90.46% FECR at 2.0 g/kg on day 15	Veterinary extract result; active terpene and human dose unresolved
A. absinthium / *T. spiralis*	Thujone-free population oils: 72-100% reduction in larval infectivity ex vivo and 66% adult reduction in vivo; no effect on *M. javanica*	Chemotype- and parasite-specific oil phenotype; not a universal wormwood effect
A. absinthium / *H. contortus*	Aqueous extract and dried-plant lamb study: ED50 1.40 and ED99 3.76 mg/mL in vitro, but no significant group-level FECR	In-vitro ovicidal activity did not translate to the tested group-level outcome
A. absinthium / *S. mansoni*	Dichloromethane extract/fractions: 100% adult-worm mortality at 200 microgram/mL; artemetin and hydroxypelenolide isolated	Extract/fraction signal; isolated-compound potency, selectivity, and host efficacy unresolved
A. absinthium / *L. amazonensis*	Essential oil in nanocochleates: intracellular-amastigote IC50 21.5 +/- 2.5 versus macrophage IC50 27.7 +/- 5.6 microgram/mL; approximately 50% murine suppression at 30 mg/kg intralesional	Formulation- and route-specific result; not evidence for a single terpene or human vermifuge

The wormwood literature therefore supports a focused experimental program. First, voucher-linked *A. absinthium* accessions should be chemically profiled for thujones, camphor, cineole, epoxyocimenes, chrysanthenol, sesquiterpene lactones, and nonvolatile co-metabolites. Second, the same preparations should be tested against matched life stages of *H. contortus* and *T. spiralis*, with adult-worm motility, egg hatch, larval development, intracellular or tissue stages where relevant, and mammalian-cell counterscreens. Third, oils and extracts should be fractionated and recombined at measured ratios to distinguish single-compound activity from matrix effects. Finally, parasite-protein interaction work should begin from the observed *A. absinthium* chemistry and parasite phenotype; docking to a convenient protein without chemical provenance is not a target discovery result.

Wormwood also sharpens the safety boundary. Thujone is a neuroactive, preparation-dependent risk marker, and some of the strongest nematode evidence above came from thujone-free cultivated oils, whereas other wormwood chemotypes may contain thujones. Consequently, "vermifuge" cannot be used as a synonym for "safe," and veterinary efficacy cannot be transferred to human self-treatment. The Terpedia records preserve the distinction among traditional use, in-vitro nematode phenotype, in-vivo veterinary efficacy, quantified oil chemistry, and human therapeutic evidence.

Safety and translation

Thujone is neurotoxic and must be tracked as a quantified, preparation-specific analyte. Artemisinin-derived antimalarial treatment should not be conflated with crude *A. annua* tea, essential oil, or an unidentified *Artemisia* preparation. In-vitro antioxidant, antimicrobial, insecticidal, cytotoxic, or antiparasitic results do not establish a safe human dose or therapeutic efficacy.

Composition-aware toxicology is necessary because risk markers vary among accessions and processing methods. A comparative toxicological study of *A. annua* volatiles paired chemical profiles with pharmacological, microbiological, and toxicity assays, demonstrating that "*A. annua* oil" is not interchangeable with artemisinin or with a standardized extract (Radulović et al. 2013). Regional *A. afra* oils likewise differed in camphor-, cineole-, and thujone-associated composition, making provenance relevant to safety interpretation (Oyedeji et al. 2009). A safety table should therefore include not only activity and a generic cytotoxicity result but quantified hazardous constituents, route, exposure duration, metabolic activation, and relevant host or ecological controls.

Preparation matrices also change exposure. Aqueous infusions extract a different chemical subset from hydrodistilled oils; solvent extracts can concentrate nonvolatile lactones and other electrophilic metabolites; and nanoformulations can alter stability, tissue distribution, and apparent potency. These are not minor formulation details. They determine which compounds reach a parasite, which host proteins are exposed, and whether an in-vitro concentration can plausibly be achieved safely in vivo.

The regulatory boundary is preparation-specific. The EMA monograph requires the amount of thujone in covered *A. absinthium* products to be specified and sets a daily exposure limit of 6.0 mg (EMA monograph). For malaria, WHO recommends quality-assured artemisinin-based combination therapy and explicitly does not support promotion of *Artemisia* plant material such as teas, tablets, or capsules for prevention or treatment (WHO treatment guidance). These authorities address defined products and clinical policy; they do not establish the safety or efficacy of an arbitrary species, oil, extract, or terpene mixture.

The record-level safety and translation boundaries are archived in `safety-translation.csv`. This prevents a low-toxicity result in one assay or a traditional-use claim from being converted into a generalized therapeutic recommendation.

Testable hypotheses and research agenda

We propose four hypotheses: (1) WGD- and lineage-specific TPS retention predicts chemical diversity only when copy number is paired with expression and metabolomics; (2) antiparasitic chemistry associated with artemisinin-like products tracks pathway completion more closely than total TPS count; (3) essential-oil activity is often a chemotype-specific mixture property; and (4) secretory-tissue expression predicts accumulated compounds better than bulk leaf expression. Discriminating tests combine frozen assemblies, standardized annotation, orthology-aware gene trees, trichome transcriptomics, targeted metabolomics, enzyme assays, and fractionated parasite assays.

Minimum reporting standard for a future meta-analysis

Each observation should be stored as a specimen-level record with a stable taxon identifier, voucher or accession, tissue, developmental stage, locality and collection date, preparation, extraction yield, analytical platform, retention-index or spectral confirmation, compound identity confidence, abundance units, and assay metadata. A quantitative synthesis should not pool normalized peak percentages with absolute concentrations, purified compounds with oils, or promastigote-only endpoints with intracellular or in-vivo endpoints. This design makes heterogeneity a modeled variable rather than an unexplained nuisance.

Evidence-audit status of this review

The article's auditable evidence is distributed across five linked layers. `sources.json` registers 133 sources; `evidence-matrix.csv` contains 117 source-linked records spanning chemistry, pathway biology, genomes, chemotypes, and parasite assays; `antiparasitic-evidence.csv` retains preparation, parasite-stage, dose, endpoint, and host-control fields; `parasite-protein-interactions.csv` separates 11 direct, functional, computational, and unresolved-target records; and `claim-audit.csv` maps 23 central article claims to source identifiers and permitted inference boundaries. The package also archives 325 bounded title/abstract screening decisions from a 1,387-record priority queue and 39 compound/specimen extraction records. These counts describe the current versioned evidence state, not a completed systematic-review denominator.

The audit supports an extensive, source-traceable synthesis but does not justify calling the review exhaustive. Manual screening, full-text eligibility assessment, quantitative harmonization, and species-tree-aware orthology remain open. Claims based on the 117-record matrix are source-linked and scope-qualified; claims based only on the unscreened queue are not used as findings. The exact package version, validation output, and SHA-256 manifest are stored in the Terpedia KB snapshot.

Conclusions

Artemisia terpene biology is best understood as a structured interaction between evolutionary history, gene-family innovation, tissue compartmentalization, ecological context, and analytical method. A publishable comparative review must therefore preserve specimen-level provenance and separate observed chemistry, direct enzyme evidence, annotation-supported hypotheses, mixture-level phenotypes, and unestablished claims. The Terpedia-linked dataset and protocol provide the basis for expanding this draft into a fully screened, quantitatively harmonized review.

Data and code availability

The manuscript protocol, source registry, evidence matrix, representative chemistry table, and Terpedia-linked pathway records are versioned with this article package. The comparative-genomics specification, completed diagnostic outputs, and exact *A. annua* and *A. argyi* protein FASTA inputs are versioned in the linked Terpedia KB study, with the large FASTAs tracked through Git LFS. Public genome and transcriptome accessions and retrieval metadata remain recorded so the archived inputs can be traced to their originating repositories. Future analyses must use frozen manifests and record software versions, parameters, checksums, and execution environment.

Core sources

See `sources.json` for the auditable registry. Key anchors include the genus essential-oil review (https://pmc.ncbi.nlm.nih.gov/articles/PMC6268508/), the *Artemisia* sesquiterpene-lactone analysis review (https://pmc.ncbi.nlm.nih.gov/articles/PMC4606394/), the *A. annua* chemotype metabolomics study (https://pmc.ncbi.nlm.nih.gov/articles/PMC5968107/), the *A. annua* genome study (https://doi.org/10.1016/j.molp.2018.03.015), the *A. argyi* chromosome-scale genome (https://pmc.ncbi.nlm.nih.gov/articles/PMC10203441/), the recent *A. argyi* comparison (https://pmc.ncbi.nlm.nih.gov/articles/PMC12702566/), the broad *Artemisia* phylogenomics framework (https://pmc.ncbi.nlm.nih.gov/articles/PMC12508166/), and the *Leishmania* assay studies (https://pmc.ncbi.nlm.nih.gov/articles/PMC4243575/; https://pmc.ncbi.nlm.nih.gov/articles/PMC5078739/; https://pubmed.ncbi.nlm.nih.gov/20397218/).